

Student Test Platform

A simple guide to how the application works

Documentation · Generated 2026-09-16

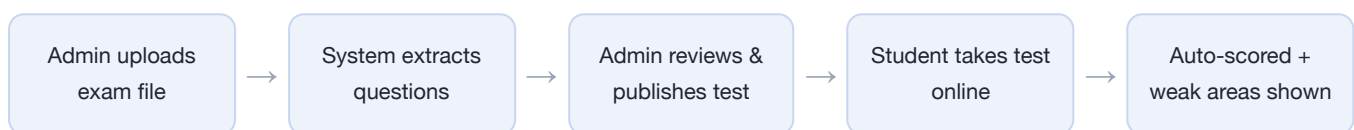
1. Purpose

This platform replaces the traditional **paper-based exam**. Instead of printing question booklets and OMR sheets and checking each one by hand, an admin simply uploads the exam content as a Word document, and the system takes care of extracting the questions, running the timed test, scoring it, and analyzing the results.

Two things this app is built to do:

1. Cut out the paperwork — no printing, no OMR sheets, no manual checking.
2. Show, clearly and instantly, **where each student is weak** — topic by topic, question by question — so that weak areas can be targeted before the real exam.

The Big Picture



Two Test Modes

Mode	What it is
Sectional	A focused practice test on one topic (e.g. just DILR). Students first look through the questions and choose which ones to attempt, then answer only those. Used for regular, topic-wise practice.
Mock	A full-length exam that simulates the real test (CAT, IPMAT and similar), with multiple timed sections. Students go straight into answering — there's no selection step — just like sitting the actual exam.

After either mode, the admin's results screen breaks performance down by student, by topic, and by question — showing exactly which concepts a student got wrong, the same insight a paper exam would otherwise take manual grading and tallying to produce.

Tech Stack

Layer	Technology
Frontend (what users see)	React + Vite, TypeScript
Backend (server & logic)	Node.js + Express, with secure login (JWT)
Database	PostgreSQL
File handling	Handles uploads, image files, and Excel exports
Document processing	A separate Python tool that reads Word files and pulls out the questions
Image text reading (optional)	Built-in OCR for questions where an answer option is a picture instead of text

2. Admin Flow

This is everything the admin does, from uploading an exam to seeing the results.

1. Upload — Admin uploads a Word (.docx) exam file. The system reads it and automatically pulls out every question, option, and correct answer.

2. Review & fix — Admin checks the extracted questions and fixes anything the automatic extraction got wrong (a stray character, a missing option, etc.). Every fix is kept on record.

3. Add questions manually (optional) — Admin can also type in questions by hand, or build a question bank from scratch — useful for mock exams.

4. Build the test — Admin picks which reviewed questions go into a test, and sets the rules: how long students get to read, how long to answer, and how many questions they need to choose.

5. Publish — Once every question is checked and approved, the admin publishes the test so students can see and take it.

6. Run the live test — Students gather in a waiting room; the admin opens the gate and everyone starts together, at the same time. The admin can also end the test early if needed.

7. View results — After the test, the admin sees a leaderboard and a breakdown of who got which questions right or wrong, and can export it all to Excel.

3. Student Flow

This is what a student experiences, from logging in to seeing their result.

1. Log in — Student signs in with their username and password.

2. Pick a test — Student sees a list of available tests and chooses one to take.

3. Wait for the start — If the test hasn't started yet, the student waits in a lobby with other students until the admin starts it for everyone at once.

4. Reading phase — Student looks through all the questions and picks which ones they want to attempt, without answering yet (sectional tests only — mock exams skip straight to answering).

5. Lock in choices — Once the student confirms their picks (or the reading time runs out), those questions are locked in and the answering timer starts.

6. Answering phase — Student answers only the questions they picked. Each answer is scored the moment it's submitted.

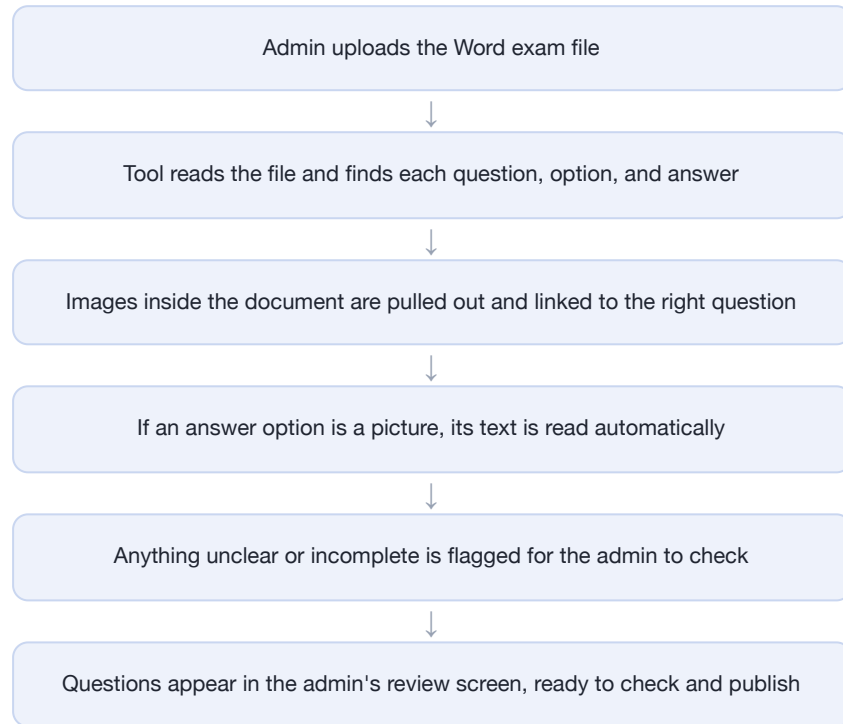
7. Submit — Student submits when done, or the test auto-submits when time runs out — so no one ever gets stuck.

8. See results — Student sees their score, which questions they got right or wrong, and how they rank against others.

Always on time. The test timers run automatically in the background. Whether a student submits manually or simply runs out of time, the system always closes the test out cleanly and scores what was answered.

4. How Exam Files Get Turned Into Questions

When an admin uploads a Word document, a separate, dedicated tool takes over to read it and pull out the questions — this is what removes the need for anyone to manually retype an exam into the system.



Why this matters. A 40-page exam paper that would normally take hours to retype into a system is turned into ready-to-review questions in seconds — the admin only needs to double-check the results, not type everything from scratch.

5. What This Replaces

Paper-based exam	This platform
Print question booklets and OMR sheets	Upload one Word file
Students fill answers by hand	Students answer on screen, on their own device
Staff manually check and tally every sheet	Answers are scored automatically the moment they're submitted
Results take days to compile	Results and rankings are ready as soon as the test ends
No easy way to see topic-wise weaknesses	Clear topic-by-topic, question-by-question breakdown of where each student is weak